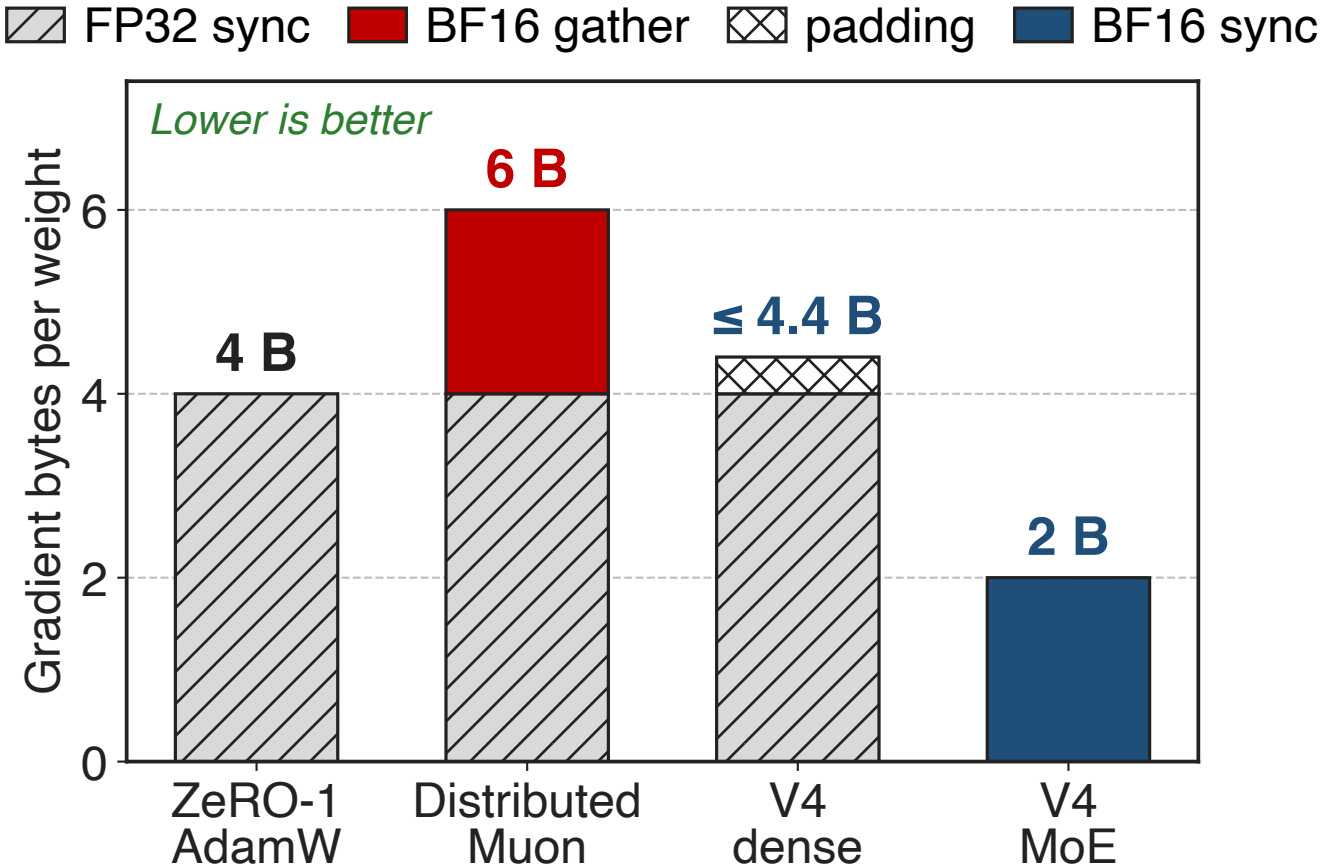
FP32 sync
BF16 gather
padding
BF16 sync
Lower is better
Gradient bytes per weight
0
2
4
6
4 B
6 B
≤ 4.4 B
2 B
ZeRO-1
AdamW
Distributed
Muon
V4
dense
V4
MoE